La Thi Ngoc Khanh

(223) 307-1430 | khanh.ltn7@gmail.com | LinkedIn | Github

Mathematics and Computer Science student with hands-on experience in ML model development, statistical modeling and data analysis. Seeking machine learning and data analysis related opportunities

EDUCATION

Franklin and Marshall College | GPA 3.75 Lancaster, PA
B.A. Candidate in Mathematics and Computer Science Expected June 2029
Relevant Coursework: Multivariable Calculus, Linear Algebra, Data Structure & Algorithm with Java, Introduction to Higher Mathematics, Machine Learning Foundations (Cornell Break Through Tech)
Organizations: ColorStack, Break Through Tech, F&M Varsity Women Crew

TECH STACKS

Languages: Python, TypeScript/JavaScript, Java, Bash
Frameworks & Libraries: scikit-learn, pandas, NumPy, Matplotlib, React, Next.js, LangChain/LangGraph, ChromaDB, Pydantic
Tools: Git/GitHub, VS Code, JupyterLab, LaTeX, Claude Code, Shapr3D

EXPERIENCE

MediMate Ithaca, NY (Remote)
Break Through Tech AI Studio Fellow Apr 2026 - Current
As part of a five-person team in a semester-long fellowship, I contribute to making an ML model to help MediMate Foundation predict the outcome of a kidney-transplant waitlist registration

- Using the filtered cohort of 494,862 kidney transplant registrations from the OPTN KIDPAN dataset, I checked data quality of the set and its outcome drift before vs after 2021, eliminating fields prone to leakages before setting up the prediction model
- I help train and fine-tune classification models to predict kidney waitlist registration outcomes
- Across demographic subgroups, I audit model fairness to flag bias before the tool informs any real decision
- By flagging at-risk registrations early, we want to give medical practitioners a data-driven signal to support transplant decisions

Power Packs Project Lancaster, PA
Programming Assistant Intern Jun 2026 – Jul 2026
On a two-person student team, I improved a cost-prediction pipeline and contributed to a meal-scheduling dashboard for the Power Packs Project, a Lancaster-based nonprofit food-distribution program

- I built multiple prediction models in Python to estimate recipe costs for Power Packs across 3, 6, and 12-month horizons, standardizing six years of data across 866 ingredients using cosine similarity, sentence-transformer embeddings, and Bayesian shrinkage
- By autonomously predicting recipe costs over time, the models saved staff an estimated 8 hours a month on meal scheduling and budgeting for donor funding requests
- The team's methodology was then documented into reusable Jupyter notebooks for future researchers and rewritten into a research paper now published in an INFORMS journal

Origami Mathematics Research Lancaster, PA
Mathematics Research Assistant Sep 2026 – Current
As part of a three-person research team, looking into an open problem in rigid origami, I investigated whether real-world, thick-material origami models can still fold flat

- I survey existing flat-foldability literature to identify and test candidate folding methods
- I build a prototype for each model using the following three stages: cardboard-and-paper fold test, then precise Shapr3D model and a 3D-printed physical part with different materials for behavioural tests
- Co-authoring a journal article on thick-material rigid origami and preparing a conference presentation of the team's findings

PROJECTS

Monet | Github

- I developed a forensic tool that helps artists prove ownership of their work and catch tampering or theft of their media
- To detect tampering, my teammate and I embedded invisible digital signatures using SHA-256 hashing and

byte-level encoding, with a Gemini layer that turns hash discrepancies into plain-language forensic reports
- This project reached the finals of the Frostbyte Hackathon

Cookie Eater | Github
- I built a Chrome extension that scores Terms of Service and privacy agreements on a 0–100 scale, flags specific risks like data selling or forced arbitration, and gives users a plain-language summary so they know what they are agreeing to
- To catch dynamically injected consent popups in real time, I used a MutationObserver paired with a heuristic scoring classifier
- I wrote a function to pull full policy text straight from the DOM, then routed it through an abstract AI service layer on a FastAPI backend that autonomously swaps between models to translate or analyze the fetched TOS